

Task 1.

The dataset contains 200 firms across Construction, Manufacturing and Transportation. All core variables (Waste, Recycle, OPEX, Empl, CAPEX, Sust_Inv) contain no missing or negative values, and all fall within plausible operational ranges. Table 1 summarises their distributions and already suggests heavy right-skewness in the main size and cost measures, particularly Waste and OPEX.

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
Waste	0.21	17.46	79.17	139.40	154.55	1670.48
Waste_red	0.0	0.0	0.0	0.4	1.0	1.0
Recycle	0.080	8.117	30.490	62.774	67.120	936.310
Green_In	0.000	0.000	1.000	0.655	1.000	1.000
Empl	1.0	10.0	61.5	151.2	175.2	1314.0
CAPEX	1.25	11.26	82.85	149.04	177.79	1890.59
Sust_Inv	0.780	2.755	4.495	4.963	6.155	19.520
OPEX	1.920	8.525	23.320	33.581	46.830	230.030
recycle_rate	0.03214	0.34146	0.47334	0.46497	0.57814	0.82072

Table 1. Summary statistics table

The main cost and size variables, Waste, OPEX, Empl and CAPEX, are strongly right-skewed. For instance, mean Waste is 139.40 tonnes, almost twice the median of 79.17 tonnes, and OPEX shows a similar spread (mean 33.58 vs median 23.32). A small number of large firms dominate the upper tail, which directly motivates the later use of log transformations. Scatterplots confirm clear non-linear scale relationships: OPEX increases disproportionately with firm size, and its variance widens among larger firms (Figure 3). This indicates heterogeneity in scale and cost structures that simple level-level models cannot capture. This pattern reminds of the example of right-skewed economic variables where scale effects generate multiplicative relationships in levels,

motivating transformations to stabilise variance and reduce heteroskedasticity before model building.

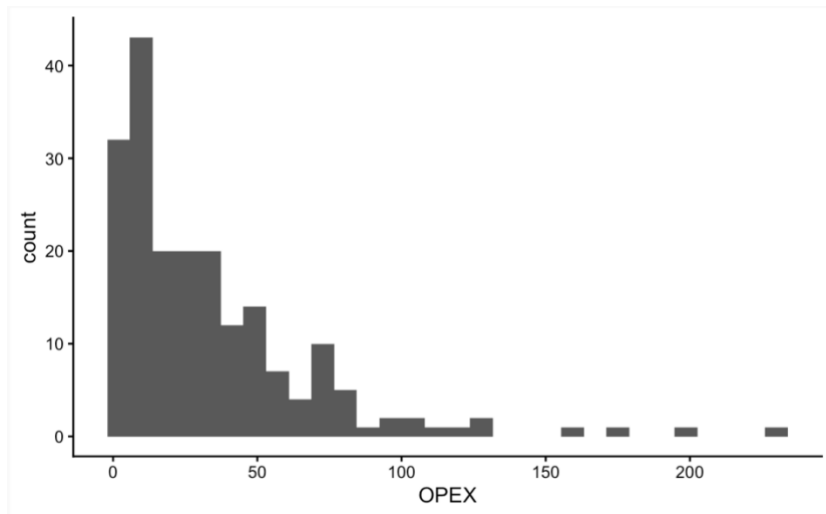


Figure 1. Histogram of OPEX

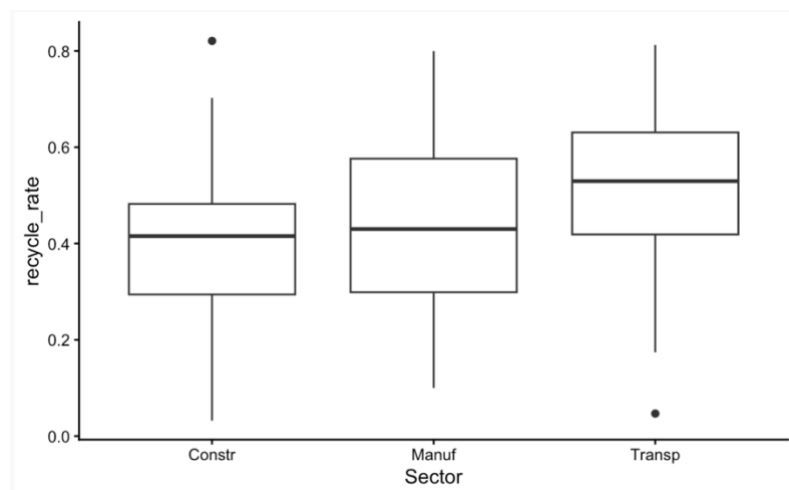


Figure 2. Boxplot of recycle rate by sector

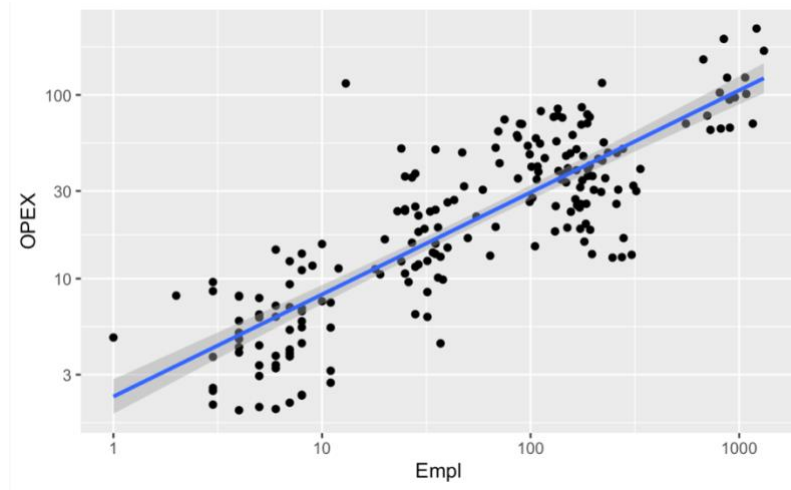


Figure 3. Scatterplot of OPEX vs Empl

Recycling behaviour, measured as Recycle/Waste, is more concentrated. The mean recycling rate is 0.465, with half of firms between roughly 0.34 and 0.58. While the range is wide (0.032 to 0.821), most firms cluster in the middle, making the ratio a more informative performance measure than raw tonnages. Because the recycling rate is scale-free by construction, it avoids the interpretational distortions caused by comparing raw Waste or Recycle volumes across firms of vastly different size.

Sectoral differences are clear. Transportation firms recycle the highest share of waste on average (0.531), followed by Manufacturing (0.441) and Construction (0.391). Construction also records the highest mean OPEX (44.34m EUR) despite the lowest employment levels, pointing to sector-specific cost structures rather than scale alone. Manufacturing and Transportation have similar average OPEX (around 29m EUR) but differ markedly in capital intensity.

	Waste	Recycle	OPEX	Empl	CAPEX	Sust_Inv
Waste	1.000	0.925	0.579	0.755	0.610	0.471
Recycle	0.925	1.000	0.454	0.693	0.603	0.475
OPEX	0.579	0.454	1.000	0.756	0.717	0.537
Empl	0.755	0.693	0.756	1.000	0.802	0.656
CAPEX	0.610	0.603	0.717	0.802	1.000	0.759
Sust_Inv	0.471	0.475	0.537	0.656	0.759	1.000

Table 2. Correlation table

Pairwise inspections (e.g., scatterplots with smoothing lines) also show that log-transformed relationships appear more linear than level-level ones, which directly informs the specification choices in subsequent regression analysis.

Overall, the EDA highlights three main features: (i) strong right-skewness in scale variables, (ii) clear sectoral clustering in cost and recycling performance, and (iii) substantial heterogeneity in firm size.

Task 2.

To identify firms with strong recycling performance, a high-performer is defined as one with a recycling rate (Recycle/Waste) above 50%. This threshold is substantively meaningful and lands around the upper half of the sample. Using this cut-off, 87 out of 200 firms qualify as high performers. This threshold also ensures roughly balanced group sizes, which improves the reliability of the chi-square test by avoiding sparse cells.

Initiative adoption was analysed using 2x2 contingency tables for high-performing and non-high-performing firms. For the Waste Reduction Initiative, all expected cell counts were well above the conventional threshold of five (e.g., $E_{\text{high-perf, WR1}} = 34.8$; $E_{\text{high-perf, WR0}} = 52.2$), confirming that the chi-square test of independence was appropriate for this dataset. The resulting chi-square test returned $p = 0.6002$, indicating no statistically significant difference in participation rates between high performers and the rest of the sample.

High-performers Yes/No	Participates (Yes/No)	0	1	Grand Total	Participation rate
0		66	47	113	41.6%
1		54	33	87	37.9%
Grand Total		120	80	200	
Total high performers:			87	out of	200

Figure 4. Waste reduction initiative pivot table

For the Waste Reduction Initiative, 37.9% of high performers participate, compared with 41.6% of other firms. The chi-square test here evaluates independence between two categorical variables: recycling-performance classification and initiative participation. The conditional probability of participation for high performers is therefore 0.379, meaning they are not more likely to join the initiative. Since both p-values are well above conventional significance levels, the observed differences in

proportions can be attributed to sampling variation rather than systematic behavioural patterns.

The Green Certification Scheme shows the same pattern. Participation among high performers is 65.5%, identical to the 65.5% rate among all other firms. The chi-square p-value of 1.000 again indicates no association. High-performing firms are therefore no more likely to be certified than their peers.

High-performer Yes/No	Participates (Yes/No)	0	1	Grand Total	Participation rate
0		39	74	113	65.5%
1		30	57	87	65.5%
Grand Total		69	131	200	

Figure 5. Green certification scheme pivot table

Taken together, the results show that neither initiative aligns with realised recycling performance. If the programmes aim to reward or incentivise stronger environmental outcomes, the lack of any link between participation and recycling rates suggests a clear design-gap. If instead the objective were to support weaker performers, higher participation among low performers – or any association with performance – would be expected, but does not appear in this cross-section. More broadly, the absence of any statistical association suggests that initiative uptake operates independently of realised environmental performance, which limits their value as performance-screening or performance-rewarding mechanisms. More effective programme design would tie eligibility or benefits to measurable recycling improvements, making resource allocation more outcome-focused.

Task 3.

To test whether OPEX differs between participants and non-participants in each initiative, two-sample unequal-variance t-tests (Welch tests) were run for each programme. The null hypothesis is equality of mean OPEX across groups. Despite OPEX being right-skewed, both groups have sample sizes above 30, so the central limit theorem supports the reliability of the test statistics. The Welch test is chosen because visual inspection and descriptive statistics suggest unequal variances across participant and non-participant groups, making the equal-variance t-test inappropriate.

For the Waste Reduction Initiative, participants record mean OPEX of 30.06m EUR compared with 35.93m EUR for non-participants, a raw difference of 5.87m EUR. The Welch test returns $p = 0.2146$, indicating no statistically significant difference. The confidence interval for the mean difference also spans zero, reinforcing the conclusion of no statistically detectable effect.

The Green Certification Scheme shows an even narrower gap: participants average 33.12m EUR and non-participants 34.46m EUR ($p = 0.7990$). In both cases, the unconditional comparisons show no systematic cost difference linked to participation.

These results reflect unconditional mean differences and therefore do not adjust for the strong size-related drivers of OPEX documented in Task 1.

However, this approach is limited. Mean comparisons cannot adjust for key confounders such as firm size, capital intensity or sector. Given the strong scale effects identified in Task 1, simple t-tests may hide structural differences. A more informative approach is a multivariate model using log-transformed OPEX, which allows participation effects to be assessed while holding firm characteristics constant. This approach assumes independence within and across groups and cannot incorporate multiple covariates simultaneously, which is why it is inappropriate for assessing partial effects when predictors are correlated.

Task 4

Using the high-performance definition from Task 2 (recycling rate $> 50\%$), the prevalence of strong recycling outcomes was compared between Manufacturing and Transportation. Manufacturing records 26 high performers out of 60 firms (43.3%), while Transportation records 52 out of 84 (61.9%).

A chi-square test of independence was applied to the 2×2 table. All expected cell counts exceeded 5, so the test assumptions were met. The chi-square statistic is 4.86 ($df = 1$), producing a p-value of 0.027. This indicates a statistically significant difference in high-performance rates at the 5% level.

The magnitude of the gap is also notable: high-performance firms are 18.6 percentage points more common in Transportation than in Manufacturing. This points to sector-level factors shaping recycling outcomes even before any further modelling.

Task 5

Model formulation and selection

The exploratory analysis in Task 1 showed that OPEX, employment (Empl), capital expenditure (CAPEX) and sustainability investments (Sust_Inv) were all strongly right-skewed. Scatterplots indicated non-linear scale patterns: larger firms displayed disproportionately higher OPEX and greater variance. Correlation results also confirmed strong positive associations between OPEX and both employment and capital expenditure, while the positive correlation between Sust_Inv and OPEX appeared to reflect size effects rather than a separate mechanism. These patterns match the multiplicative scale relationships in the log-transform lecture, where log transformations linearise curved level – level plots.

To address the skewness and stabilise variance, natural logarithms were applied to the four scale variables: log_OPEX (dependent variable), log_Empl, log_CAPEX and log_Sust_Inv. Residual – fitted plots showed improved variance stability. The initiative indicators and sector variable were treated as factors, following standard handling of categorical predictors.

An initial, broader model (model_v1) included Recycle_rate, but the variable was statistically insignificant, and did not meaningfully improve explanatory power: R^2 increases only from 0.8865 to 0.8876 and adjusted R^2 from 0.8824 to 0.8829, a change of less than 0.001). Its inclusion added complexity among the sustainability and initiative predictors. Given the negligible contribution of Recycle_rate, retaining it would violate the parsimony principle emphasised in the lectures, by adding parameters that increase variance without improving fit. Model_v2 – excluding Recycle_rate – was therefore selected as the preferred specification. It retains all theoretically relevant drivers (firm scale, sustainability intensity, initiative participation and sector effects) while remaining more parsimonious. All subsequent interpretation is based on model_v2.

Variable	Model v1	Model v2 (preferred)	95% CI (Model v2)
Intercept	1.2737 ***	1.1655 ***	[0.9783, 1.3528]
log_Empl	0.2902 ***	0.2898 ***	[0.2123, 0.3673]
log_CAPEX	0.4811 ***	0.4873 ***	[0.4013, 0.5734]
log_Sust_Inv	-0.4591 ***	-0.4653 ***	[-0.6381, -0.2925]
Recycle_rate	-0.2370	—	—
Green_In_f(1)	-0.0350	-0.0358	[-0.1532, 0.0817]
Waste_red_f(1)	-0.0597	-0.0559	[-0.1818, 0.0700]
sector_f: Manuf	-0.6060 ***	-0.6172 ***	[-0.7597, -0.4748]
sector_f: Transp	-0.7904 ***	-0.8236 ***	[-0.9589, -0.6883]
Residual Std. Error	0.3767	0.3776	
R ² / Adj. R ²	0.8876 / 0.8829	0.8865 / 0.8824	
F-statistic	188.5 ***	214.2 ***	
Notes:	*** p < 0.001; ** p < 0.01; * p < 0.05.		

Table 3. Regression results for log(OPEX)

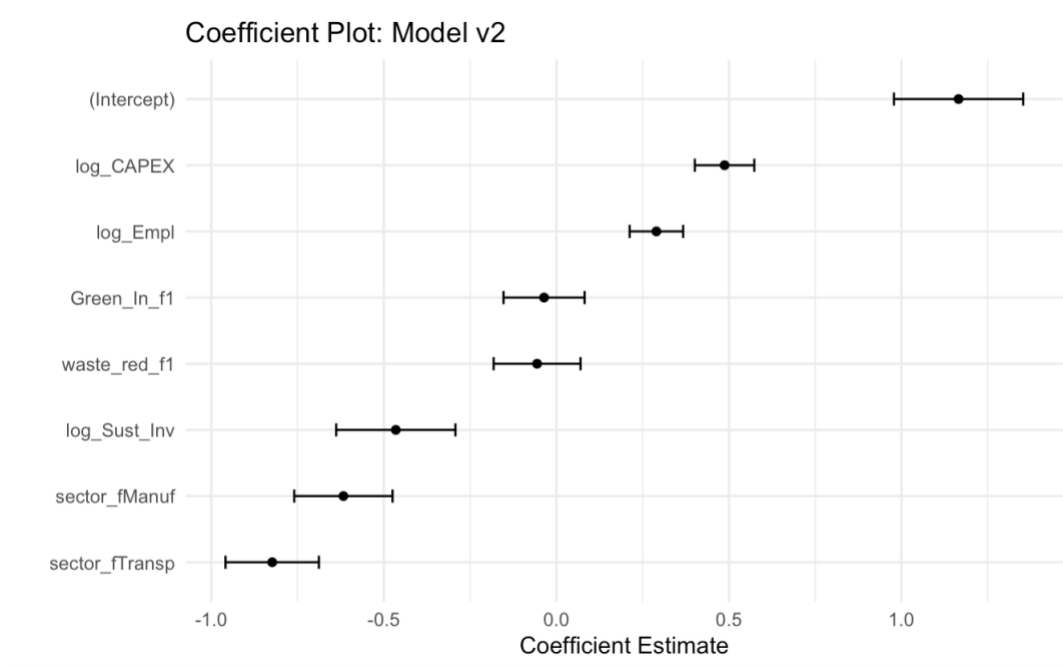


Figure 6. Coefficient Plot (Model v2)

(a) Main drivers of OPEX and their contributions

In model_v2, the three log-transformed scale variables – employment, capital expenditure and sustainability investments – are the main continuous determinants of OPEX. Their coefficients function as elasticities due to the log – log structure.

Employment has a coefficient of 0.2898 ($p < 0.001$), meaning a 1% increase in employment is associated with a 0.29% increase in OPEX. A 10% rise in employment corresponds to roughly 2.9% higher operating expenditure. Capital expenditure has a larger elasticity of 0.4873 ($p < 0.001$), implying that a 1% increase in CAPEX corresponds to a 0.49% increase in OPEX. This indicates that more capital-intensive firms carry substantially higher operating costs.

Sustainability investments move in the opposite direction. The coefficient on $\log_Sust_Inv$ is -0.4653 ($p < 0.001$). A 1% increase in sustainability expenditure is associated with a 0.47% reduction in OPEX; a 10% increase corresponds to an estimated 4.7% decrease. This negative elasticity is statistically robust and suggests that sustainability investments in this dataset relate to efficiency-enhancing practices such as waste minimisation or energy optimisation.

Sector effects are also substantial. With Construction as the baseline category, Manufacturing has a coefficient of -0.6172 and Transportation -0.8236 (both $p < 0.001$). Exponentiating these coefficients implies that Manufacturing firms have around 46% lower OPEX than comparable Construction firms, and Transportation firms around 56% lower.

These sector effects represent conditional mean shifts relative to Construction, holding all continuous predictors constant.

Overall model fit is strong: $R^2 = 0.8865$ and adjusted $R^2 = 0.8824$ show that nearly 89% of cross-firm variation in $\log(OPEX)$ is explained by the specification.

(b) Elasticity of OPEX with respect to sustainability investments

The elasticity of OPEX with respect to sustainability investment is given by the coefficient on $\log_Sust_Inv$ (-0.4653). A 1% increase in sustainability expenditure is associated with a 0.47% decrease in OPEX, holding all other variables constant. The confidence interval (-0.64 to -0.29) excludes zero, confirming statistical reliability and economic relevance.

(c) Do the initiatives significantly affect OPEX?

The initiative indicators ($Green_ln_f$ and $waste_red_f$) estimate percentage differences in OPEX between participants and non-participants. Both coefficients are

small and statistically insignificant: -0.0358 for Green_In_f ($p = 0.549$) and -0.0559 for waste_red_f ($p = 0.382$). These map to approximate differences of -3.6% and -5.6% , but the confidence intervals include zero. Once employment, capital expenditure, sustainability investment and sector are accounted for, initiative participation does not explain variation in OPEX. This aligns with the descriptive evidence from Tasks 2 and 3, which also showed no systematic cost or performance differences linked to participation.

(d) Sectoral differences in OPEX after controlling for other factors

Both sector coefficients are large, negative and highly significant. After adjusting for scale variables and sustainability investments, Manufacturing and Transportation firms operate with substantially lower OPEX than Construction firms. While the model does not directly test differences between the two non-baseline sectors, the coefficient sizes suggest that Transportation shows the lowest conditional expenditure, followed by Manufacturing. These sectoral differences are consistent with the descriptive patterns identified in Task 1. These patterns highlight persistent sector-specific cost structures.

Overview

The final model aligns with the exploratory analysis and provides a clear explanation of OPEX variation. Operating expenditure is most strongly associated with firm scale and capital intensity; sustainability investments are linked to lower operating costs; initiative participation does not meaningfully affect OPEX; and large sectoral differences remain after adjustment. The model offers high explanatory power while remaining parsimonious.

Task 6.

Diagnostic analysis

Diagnostic analysis was carried out on the preferred specification (model_v2), which models log-transformed OPEX using log – log continuous predictors and factor variables for initiatives and sector. The goal is to assess whether the core OLS assumptions hold and whether any deviations would meaningfully affect the findings reported in Task 5.

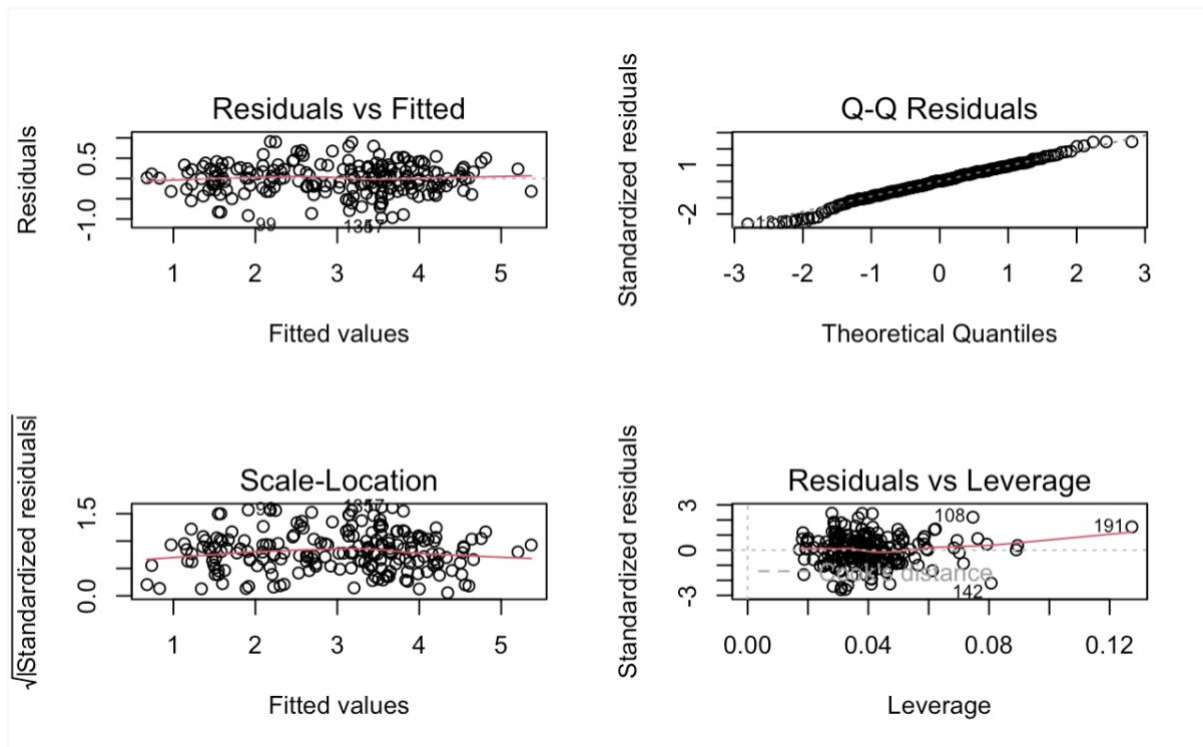


Figure 7.1-7.4. Diagnostic Analysis

Linearity and functional form

Standard diagnostic plots indicate that the log – log specification is appropriate. In the Residuals vs Fitted plot, residuals are centred around zero with no visible curvature, consistent with a linear conditional mean for $\log(\text{OPEX})$. The Scale – Location plot shows an even residual spread, suggesting that the log transformation has absorbed the scale heterogeneity present in the raw data. There is no indication of functional form problems.

Normality of residuals

Normality was assessed using the Q – Q plot and the Shapiro – Wilk test. The Q – Q plot aligns closely with the 45° line, with only mild tail deviations. The Shapiro – Wilk p-value of 0.3512 means the null of normality cannot be rejected. With 200 observations, small non-normalities are not consequential: the Central Limit Theorem ensures that coefficient estimates remain approximately normally distributed. The t-tests and confidence intervals used earlier therefore remain valid.

Homoskedasticity

The Breusch – Pagan test yields $p = 0.1328$, providing no evidence against homoskedasticity. Residual plots show only a slight increase in dispersion at higher fitted values and no strong funnel pattern. As a precaution, the model was re-estimated with heteroskedasticity-robust (HC1) standard errors. The significance of the log scale variables and sector dummies remained unchanged, and both initiative indicators stayed insignificant. This confirms that the core results do not depend on the homoskedasticity assumption.

Multicollinearity

Variance Inflation Factors (VIFs) indicate moderate multicollinearity among the log-transformed scale variables, reflecting normal co-movement of firm size measures. All VIF values are well below conventional concern thresholds (≈ 10). Importantly, the coefficients on `log_Empl`, `log_CAPEX` and `log_Sust_Inv` remain stable and precisely estimated, with unchanged signs. As noted in the lecture material, multicollinearity at this level affects efficiency rather than interpretability.

Outliers, leverage and influence

Cook's distance identifies several observations with moderate influence, which is typical for cross-sectional firm data. None exceed standard thresholds for extreme influence. Re-estimating the model without these points leaves coefficient magnitudes and significance patterns effectively unchanged. There is no indication that the results hinge on outliers or leverage points.

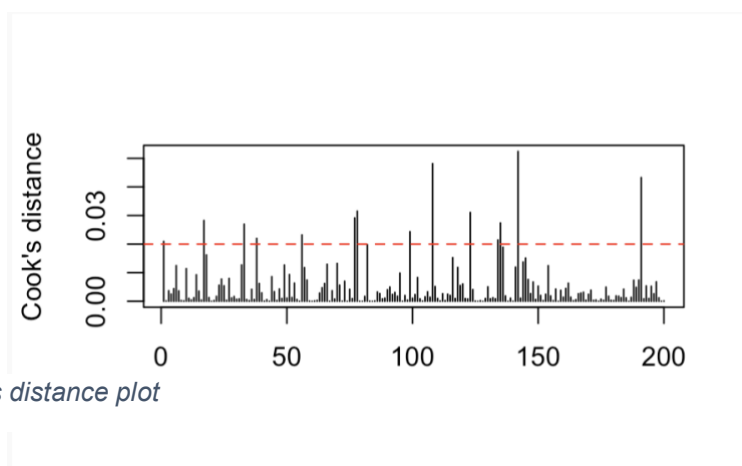


Figure 8. Cook's distance plot

Summary

Across all diagnostics: linearity, normality, homoskedasticity, multicollinearity and influence – the model meets the conditions needed for reliable OLS inference. Where

minor deviations appear, robustness checks show that the main findings are unaffected. Overall, the diagnostics support the stability and validity of the conclusions on the determinants of operating expenditure.